

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, enabling real-time attendance tracking and cutting processing time by 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with 90% coverage gates across 20+ tests.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Introduced Neo4j to offload recursive relational queries from PostgreSQL, cutting p95 latency 48% (850ms → 440ms) and eliminating SQL join overhead for multi-hop traversals.
- Refactored Cypher with relationship-type indexing; indexed traversals ran 3× faster than equivalent SQL self-joins and removed full-table scans on relationship lookups.
- Integrated dual-provider OAuth 2.0 (Google + GitHub) with httpOnly JWT cookies, refresh token rotation, and user-scoped graph access control.
- Implemented GitLab CI/CD pipeline on Linux runners automating build, test, and containerized deployment with Dockerized services and branch-protected production merges, enforcing 85% coverage across 35+ tests.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered multi-client agentic voice assistant that batches natural dialogue, resolves implicit speaker intent, and dispatches real tasks across concurrent voice streams.
- Reduced RESTful API response time by 73% by offloading RAG pipeline to async Redis/Celery task queue.
- Built bidirectional Electron + Next.js client with an embedded HTTP server for OAuth callback interception; real-time events via IPC and a dedicated webhook server deliver sub-50ms propagation.
- Deployed FastAPI inference service on AWS Lambda with lazy-loaded ML models (Whisper, Sentence-BERT, spaCy), eliminating idle compute costs while reducing cold-start latency by 86% (6,643ms → 930ms).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow* Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel where full-body movements cast spells, block attacks, and trigger abilities against an AI opponent.
- Built full-body gesture pipeline using MediaPipe Pose to map 33 skeletal landmarks into game actions; optimized inference loop to sustain 60 FPS with sub-20ms input lag.
- Developed custom 2D vector physics engine with AABB collision detection to accurately translate live 3D skeletal coordinates to game state, maintaining <5px drift at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash

Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis

Databases PostgreSQL, Neo4j, MongoDB

Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, Git, GitLab CI/CD, Linux/Unix